A Cube-Component Exit for Three-Dimensional Keller Maps

Author information withheld pending priority review

July 26, 2026

Abstract

Let k be an algebraically closed field of characteristic zero. We establish the following structural constraint: a polynomial

$$f = C + \ell^3 + Q_2 + L_1 \in k[x, y, z],$$

where $\ell \neq 0$ is linear and Q_2, L_1 are homogeneous of degrees two and one, is a polynomial coordinate whenever it has no critical point. The proof is an elementary rank classification of the quadratic form transverse to ℓ, and gives a coordinate automorphism and inverse of degree at most three. Over $\mathbf{C}$, a Keller map of degree d having such a target-linear combination of its components therefore reduces fibrewise to plane Keller maps of degree at most $3d$. Using the reported plane counterexample floor 108, this proves invertibility for $d \leq 35$; Moh's classical range gives the conservative fallback $d \leq 33$. We record only a finite, machine-checked application to a fixed denominator and make no global Jacobian or denominator-row claim.

Draft freeze (UTC): 2026-07-26T07:53:50Z.
First public release (UTC): `2026-07-26T08:13:13Z`.

1 The coordinate theorem

Recall that $f \in k[x, y, z]$ is a *coordinate* if there are $g, h \in k[x, y, z]$ with $k[f, g, h] = k[x, y, z]$. We call f *nonsingular* if its three partial derivatives have no common zero in k^3.

Theorem 1 (Cube-component coordinate theorem)**.** *Let k be algebraically closed of characteristic zero. Let ℓ be a nonzero linear form, let Q_2, L_1 be homogeneous of degrees two and one, and let $C \in k$. If*

$$f = C + \ell^3 + Q_2 + L_1$$

is nonsingular, then f is a coordinate of $k[x, y, z]$. It can be included in a polynomial coordinate system whose forward and inverse maps both have degree at most three.

Proof. A linear source change sends ℓ to x. A constant translation removes C. With $Y = (y, z)^t$, write

$$f = x^3 + Ax^2 + x\, c^t Y + \frac{1}{2} Y^t M Y + ax + d^t Y, \tag{1}$$

where $M = M^t \in M_2(k)$ and $c, d \in k^2$. Then

$$\nabla_Y f = MY + cx + d, \qquad f_x = 3x^2 + 2Ax + c^t Y + a. \tag{2}$$

We split by the rank of M.

Rank two. If M is invertible, $\nabla_Y f = 0$ has the unique solution

$$Y = -M^{-1}(cx + d).$$

After this substitution, f_x is the explicit univariate quadratic

$$3x^2 + \big(2A - c^t M^{-1} c\big)x + \big(a - c^t M^{-1} d\big). \tag{3}$$

Its leading coefficient is 3, so it has a root $x_0 \in k$. Taking $Y_0 = -M^{-1}(cx_0 + d)$ makes every derivative in (2) vanish. Thus this rank is impossible.

Rank one. By a constant linear change in Y, equation (1) becomes

$$f = g(x) + x(\beta u + \gamma v) + \frac{m}{2}u^2 + \varepsilon u + \eta v, \qquad m \neq 0. \tag{4}$$

Here $g(x) = x^3 + Ax^2 + ax$. If $\gamma \neq 0$, the explicit point

$$x_0 = -\frac{\eta}{\gamma}, \qquad u_0 = -\frac{\beta x_0 + \varepsilon}{m}, \qquad v_0 = -\frac{g'(x_0) + \beta u_0}{\gamma} \tag{5}$$

satisfies $f_v = f_u = f_x = 0$, in that order. Suppose $\gamma = 0$. If also $\eta = 0$, put

$$u(x) = -\frac{\beta x + \varepsilon}{m}.$$

Then $f_u = f_v = 0$, while

$$f_x\big(x, u(x), v\big) = 3x^2 + \left(2A - \frac{\beta^2}{m}\right)x + \left(a - \frac{\beta\varepsilon}{m}\right). \tag{6}$$

Choose a root x_0, set $u_0 = u(x_0)$, and take $v_0 = 0$. This is a critical point. Nonsingularity therefore forces $\gamma = 0, \eta \neq 0$. In this last case

$$f = \eta v + R(x, u), \qquad R = g(x) + \beta x u + \frac{m}{2}u^2 + \varepsilon u.$$

The coordinate map $(x, u, v) \mapsto (x, u, f)$ has inverse

$$(x, u, w) \longmapsto \left(x, u, \frac{w - R(x, u)}{\eta}\right). \tag{7}$$

Rank zero. Write the rank-zero expression with scalar coefficients as

$$f = g(x) + x(by + cz) + ey + hz. \tag{8}$$

Put $\Delta = bh - ce$. If $\Delta \neq 0$, use

$$u = by + cz, \qquad v = ey + hz.$$

Then

$$f = g(x) + xu + v, \qquad v = f - g(x) - xu,$$

and recover the original transverse variables by

$$y = \frac{hu - cv}{\Delta}, \qquad z = \frac{-eu + bv}{\Delta}. \tag{9}$$

Thus (x, u, f) is a coordinate system with a degree-three inverse.

Assume $\Delta = 0$. First suppose $(b, c) = (0, 0)$. If $(e, h) \neq (0, 0)$, take $v = ey + hz$ and a complementary coordinate. When $e \neq 0$, choose $u = z$, so

$$z = u, \qquad y = \frac{v - hu}{e};$$

when $e = 0$, necessarily $h \neq 0$, and one may choose $u = y$, so

$$y = u, \qquad z = \frac{v}{h}.$$

In both subcharts

$$f = g(x) + v, \qquad v = f - g(x).$$

If also $(e, h) = (0, 0)$, choose a root x_0 of g' and set $y_0 = z_0 = 0$; this is a critical point.

Finally suppose $(b, c) \neq (0, 0)$. Since $\Delta = 0$, there is $\delta \in k$ with $(e, h) = \delta(b, c)$. This assertion involves no hidden division: use $\delta = e/b$ on $b \neq 0$, and $\delta = h/c$ on the boundary $b = 0, c \neq 0$. At $x_0 = -\delta$, both transverse derivatives vanish. If $b \neq 0$, take

$$y_0 = -\frac{g'(x_0)}{b}, \qquad z_0 = 0;$$

if $b = 0, c \neq 0$, take

$$y_0 = 0, \qquad z_0 = -\frac{g'(x_0)}{c}.$$

Then $f_x = 0$ as well, giving a critical point.

The only nonsingular cases are therefore the explicit coordinate cases above. Their formulas, together with the initial linear changes, have degree at most three in both directions. $\square$

Remark 2 (Pivot completeness). *For $M = \left(\begin{smallmatrix} m_{11} & m_{12} \\ m_{12} & m_{22} \end{smallmatrix}\right)$, the rank-one chart $m_{11} \neq 0$ uses the null vector $(-m_{12}, m_{11})$; explicitly, the quadratic part is*

$$\frac{m_{11}}{2}\left(y + \frac{m_{12}}{m_{11}} z\right)^2.$$

On the boundary $m_{11} = 0 = \det M$, one has $m_{12} = 0$, and rank one is exactly $m_{22} \neq 0$. The remaining boundary is $M = 0$. In rank zero, $\det[c\ d] \neq 0$ is the independent chart. Its zero boundary splits into $c = 0, d \neq 0$, $c = d = 0$, and $c \neq 0, d = \delta c$; the last is covered by the two ordinary pivots on the components of c. Thus no coefficient is divided by without its zero locus being included.

2 The Keller exit over C

Lemma 3 (Fibre reduction). *Let $F : \mathbf{C}^3 \to \mathbf{C}^3$ have constant nonzero Jacobian and degree d. Suppose one component of F is a coordinate belonging to a coordinate automorphism whose inverse has degree at most e. If every complex plane Keller map of degree at most ed is an automorphism, then F is an automorphism.*

Proof. Permute the target components and precompose by the coordinate inverse to obtain

$$G = (G_1(u, v, w), G_2(u, v, w), w).$$

Its Jacobian determinant is a nonzero constant, and expansion along the last row gives

$$\frac{\partial(G_1, G_2)}{\partial(u, v)} \in \mathbf{C}^\times.$$

Moreover, $\deg G_i \leq ed$. For every $w_0 \in \mathbf{C}$, the specialized pair $(G_1(u, v, w_0), G_2(u, v, w_0))$ is therefore a plane Keller map in the assumed range and hence an automorphism. Thus G is injective fibre by fibre. Ax–Grothendieck implies that the injective polynomial self-map G of $\mathbf{A}^3_{\mathbf{C}}$ is surjective. The Keller condition makes G étale. An injective étale morphism over an algebraically closed field is universally injective and hence an open immersion [7]; surjectivity makes that open immersion an isomorphism. Thus G, and therefore F, is an automorphism. □

Corollary 4 (Degree 35, using the plane floor 108)**.** *Let $F : \mathbf{C}^3 \to \mathbf{C}^3$ be a Keller map of degree d. Suppose there is a nonzero target covector $\alpha \in (\mathbf{C}^3)^*$ such that*

$$\alpha{\cdot}F = C + \ell^3 + Q_2 + L_1$$

with $\ell \neq 0$ linear. If the plane Keller conclusion is valid for maximum degree < 108, then F is an automorphism for $d \leq 35$.

Proof. Extend α to a matrix $T \in \mathrm{GL}_3(\mathbf{C})$, and replace F by TF, placing αF in the third component. The new determinant is

$$\det J(TF) = (\det T)(\det JF) \in \mathbf{C}^\times.$$

Consequently the gradient of $\alpha{\cdot}F$, now the third row of an everywhere invertible Jacobian matrix, is nowhere zero. Apply Theorem 1 with $e = 3$, then Lemma 3. For $d \leq 35$,

$$ed \leq 3 \cdot 35 = 105 < 108.$$

□

The plane input in Corollary 4 is the lower floor reported in [3]. We verified the numerical claim in the primary preprint; a peer-reviewed journal version was not located.

Corollary 5 (Conservative Moh fallback)**.** *Under the hypotheses of Corollary 4, F is an automorphism for $d \leq 33$ using only the classical plane range < 100.*

Proof. Here $3d \leq 99 < 100$, so the same fibre argument applies with Moh's low-degree result [5]. □

3 Finite denominator application

The theorem is applied only to the entries in Table 1. The table is machine-checked against a frozen denominator containing exactly 26 families.

The generic `D3-SF-20C` normal form is

$$R = X^2((5 - 3z)p + 4rq).$$

Accordingly, neither that whole family nor its reciprocal $z = 1/3$ sheet is claimed. No quartic row, other denominator family, or global degree bound follows from this finite bridge. The three newly excluded whole families move the frozen fine count from 6/26 to 9/26. The containing row remains open, the global quartic status remains 4/14 certified, 4/14 provisional, and 6/14 open, and the universal total-degree floor remains four.

Identifier	Cube component	Status here
PF-BRANCH-FOURTH-THIRD	$R = p^3$	newly excluded
D3-BB-30	$R = p^3$	newly excluded
D3-OB-300	$R = p^3$	newly excluded
D4-DN-3	$R = L^3$	already excluded
D3-SF-20C, $z = 3$ only	$R = X^3$	newly excluded pivot only

Table 1: Exact scope of the frozen denominator bridge.

4 Reproducibility and priority

The dependency-free exact verifier checks the critical-point witnesses, all coordinate inverses, the degree boundaries

$$3 \cdot 35 = 105 < 108, \quad 3 \cdot 36 = 108, \quad 3 \cdot 33 = 99 < 100,$$

and the exact denominator bridge. The primary strict wrapper `verify_strict.sh` requires deliberate mutations of each of these claims to fail and prints `CUBE_COMPONENT_EXIT_STRICT_PASS`. The aggregate wrapper `verify_all_strict.sh` additionally requires the independently implemented hostile audit and prints `CUBE_COMPONENT_ALL_STRICT_PASS` only after both suites pass.

The precise coordinate lemma was not located in the limited literature search accompanying this draft. Related work includes classifications of cubic polynomials at infinity [6], broad criteria for polynomial variables [4], Vistoli's theorem for entire degree-three Keller maps [8], and the classification of degree-three automorphisms by Blanc and van Santen [2]. None was found to state the present component criterion. Worldwide novelty and priority remain unresolved.

Status and disclosure

This is an AI-assisted research draft and has not been peer reviewed. The exact primary and hostile verification suites check the rank atlas, inverse formulas, degree arithmetic, and frozen-data bridge encoded in the accompanying scripts. Those checks are evidence about the encoded algebra, not peer review, and do not mechanically verify the external plane-degree theorem, Ax–Grothendieck, or worldwide priority. The separate hostile algebra audit and hypothesis/reference audit are included in the release directory.

References

[1] J. Ax, Injective endomorphisms of varieties and schemes, *Pacific J. Math.* 31 (1969), 1–7.

[2] J. Blanc and I. van Santen, Automorphisms of the affine 3-space of degree 3, *Indiana Univ. Math. J.* 71 (2022), 857–912.

[3] J. A. Guccione, J. J. Guccione, R. Horruitiner, and C. Valqui, Increasing the degree of a possible counterexample to the Jacobian Conjecture from 100 to 108, arXiv:2204.14178 (2022), `https://arxiv.org/abs/2204.14178`.

[4] S. Kaliman, Polynomials with general $\mathbf{C}^2$-fibers are variables. I, arXiv:math/9912058, `https://arxiv.org/abs/math/9912058`.

[5] T. T. Moh, On the Jacobian conjecture and the configurations of roots, *J. Reine Angew. Math.* 340 (1983), 140–212.

[6] N. R. Ribeiro, Classification at infinity of degree-three polynomials in three variables, arXiv:2201.11026, `https://arxiv.org/abs/2201.11026`.

[7] The Stacks Project Authors, Étale and universally injective morphisms are open immersions, Tag 025G, `https://stacks.math.columbia.edu/tag/025G`.

[8] A. Vistoli, The Jacobian conjecture in dimension 3 and degree 3, *J. Pure Appl. Algebra* 142 (1999), 79–89.